Exercises

Algebraic problems

Question 3.1: Use the chain rule to differentiate the following functions:

(a) $y = \exp(x^2 + 5x + 2)$

(b) $y = \cos(3x + 5)$

(c) $y = \sin(x)^2$

Question 3.2: Use the product rule to differentiate the following functions:

(a) $y = x\exp(x)$

(b) $y = x^2\exp(4x)$

(c) $y = (x^2 + 9)\exp(-x)$

Question 3.3: Use the quotient rule to differentiate the following functions:

(a) $y = \frac{\exp(-x)}{x}$

(b) $y = \frac{x}{x^2+25}$

(c) $y = \frac{x^2}{x^3+25x}$

Question 3.4: Use the chain rule for (a) and the result of (a) in (b)-(e) find $\frac{dy}{dx}$ of the following equations (Hint: you may need to use a mixture of chain, product and quotient rules):

(a) $y = \sqrt{x^2 + 25}$

(b) $y = \sin\sqrt{x^2 + 25}$

(c) $y = x\sqrt{x^2 + 25}$

(d) $y = \frac{x}{\sqrt{x^2+25}}$

Question 3.5: Find where the following functions are decreasing and increasing. These problems require a number of steps. We know that where the derivative is 0 it is likely the function will be changing direction either side. We will come across the important property a zero derivative has soon. But for now, once you find where the derivatives are 0 you can then explore regions either side to determine if the gradient is positive or negative.

(a) $V(t) = (4 - t^2)(1 + 5t^2)$

(b) $f(x) = \frac{x-x^2}{1+8x^2}$

Question 3.6: Use the chain, product or quotient rules to find the derivative of the given function.

(a) $Z(s) = (\sin(x^2))^2$

(b) $S(x) = (1 + \sqrt{x^3})(x^{-3} - 2\sqrt[3]{x})$
(c) $R(w) = \frac{3w+w^4}{2w^2+1}$

Real-world problems

Question 3.7: Imagine we are studying the volume of water in an ecosystem reservoir over time (in hours).

(a) If we say that the volume of water in the sink can be described as:

$$V(t) = V_0 \exp\left(-\frac{t^2}{600}\right)$$

Where V_0 is the volume of water in our system when $t = 0$, in m^3. Use the chain rule to determine the gradient analytically. What is the drainage rate after 10 hours when $V_0 = 10$?

(b) If we say that the volume of water in the sink can be described as:

$$V(t) = \frac{V_0(t+10)}{10} \exp\left(-\frac{t}{5}\right)$$

Where V_0 is the volume of water in our system when $t = 0$, in m^3. Use the product rule to determine the gradient analytically. What is the drainage rate after 10 hours when $V_0 = 10$?

(c) If we say that the volume of water in the sink can be described as:

$$V(t) = \frac{V_0(x+10)}{(x^2+10)}$$

Where V_0 is the volume of water in our system when $t = 0$, in m^3. Use the quotient rule to determine the gradient analytically. What is the drainage rate after 10 hours when $V_0 = 10$?